Output tables for 1xN statistical comparisons.

March 10, 2025

1 Average rankings of Friedman test

Average ranks obtained by each method in the Friedman test.

Algorithm	Ranking
Random	6.625
NN	4.75
Sweep	6.375
SaveSeq	1.25
SaveParall	3.5
MJ	2.125
CMT	3.375

Table 1: Average Rankings of the algorithms (Friedman)

Friedman statistic (distributed according to chi-square with 6 degrees of freedom): 42.535714.
P-value computed by Friedman Test: 0.

2 Post hoc comparison (Friedman)

P-values obtained in by applying post hoc methods over the results of Friedman procedure.

i	algorithm	$z = (R_0 - R_i)/SE$	p	Holm	Finner	Li
6	Random	4.976283	0.000001	0.008333	0.008512	0.030638
5	Sweep	4.744828	0.000002	0.01	0.016952	0.030638
4	NN	3.24037	0.001194	0.0125	0.025321	0.030638
3	SaveParall	2.083095	0.037243	0.016667	0.033617	0.030638
2	CMT	1.967368	0.049141	0.025	0.041844	0.030638
1	MJ	0.810093	0.417887	0.05	0.05	0.05

Table 2: Post Hoc comparison Table for $\alpha = 0.05$ (FRIEDMAN)

Holm's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.016667.
Finner's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.033617.
Li's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.030638.

3 Adjusted P-Values (Friedman)

Adjusted P-values obtained through the application of the post hoc methods (Friedman).

i	algorithm	unadjusted p	p_{Holm}
1	Random	0.000001	0.000004
2	Sweep	0.000002	0.00001
3	NN	0.001194	0.004775
4	SaveParall	0.037243	0.111728
5	CMT	0.049141	0.111728
6	MJ	0.417887	0.417887

Table 3: Adjusted p-values (FRIEDMAN) (I)

i	algorithm	unadjusted p	p_{Finner}	p_{Li}
1	Random	0.000001	0.000004	0.000001
2	Sweep	0.000002	0.000006	0.000004
3	NN	0.001194	0.002386	0.002047
4	SaveParall	0.037243	0.05534	0.060131
5	CMT	0.049141	0.058675	0.077846
6	MJ	0.417887	0.417887	0.417887

Table 4: Adjusted p-values (FRIEDMAN) (II)